

Slide No: 1

The image shows a presentation slide with a dark blue background featuring a white grid pattern. On the left side, there is a vertical teal-colored bar. The text is as follows:
- Top right: "Topics on Electrical Engineering" in white.
- Center: "B.Sc" in large white font, followed by "Drive Technology" in a slightly smaller white font, and "LectureSlide" in a smaller white font below it.
- Bottom left: "D. T. McGuinness, Ph.D" in white.
- Bottom right: "SS.2025" in white.
- Bottom right corner: A colorful geometric logo consisting of several overlapping triangles in shades of blue, orange, and yellow.

Introduction

Slide No: 3

Slide No: 2

Table of Contents



1. Introduction
2. Magnetic Circuits and Materials

[illegible]

Table of Contents	
Table of Contents	
First Steps	
Introduction	Requirement and Learning Outcomes
Lecture Contents	Lecture Information
	Assignments
	Lecture Sources
	Content Preview

Slide No: 4

Slide No: 5

Introduction



- The goal of this lecture is to give an introduction to the fascinating world of electric machines.
- We will start from understanding how they work to the types currently used in commercial and industrial applications.
- The lecture will focus on mathematical and engineering concepts which will have direct use with real-life engineering applications.
- This lecture is a total of 3 SWS with a total of forty-five (45) UE.
- A unit (UE) is defined as 45 min lecture.

First Steps

Introduction

Introduction



- Lecture materials and all possible supplements will be present in its Github Repo.
 - You can easily access the link to the web-page from [here](#).

Github is chosen for easy access to material management and CI/CD capabilities and allowing hosting websites.

- In the lecture content is also distributed as a WebBook which can be accessed from the [Repo website](#).

First Steps

Lecture Contents

Slide No: 7

Slide No: 6

Introduction



- For two (2) sessions there will be a self-guided study which will require you to work through the material on your own.
- In these sessions you should focus on two topics: **Electromechanic Energy conversion** and **Rotating Magnetic Fields**. You can either access this content on WebBook or LectureBook which hold the **same** content.
- The next face-to-face session will focus on a recap of these topics along with any questions you may have. Afterwards there will be a 15 min long quiz covering these two topics **only**.

First Steps

Introduction

Introduction



- The student should have a good background in circuit analysis and familiarity with integral/differential calculus.
- After completion, the student will have a good grasp of electrical machines from their construction, how they are designed, what types out there and what applications are possible.

Requirements	Taught Lecture	Code	Degree	Outcome
Vector Calculus	Mathematik 1	MAT	B.Sc	Python Programming
Differential Calculus	Mathematik 2	MAT	B.Sc	Electric Drive Fundamentals
Linear Algebra	Mathematik 1	MAT	B.Sc	Magnetic Circuits
Physics	Physik	PHY		Electric Motor Analysis
-				-

Table 1: Distribution of materials across the semester.

First Steps

Requirement and Learning Outcomes

Slide No: 8

[illegible]

Slide No: 9

Description	Value
Official Name	Antriebstechnik
Lecture Code	ANT
Module Code	MECH-B-5-ANT-ANT-ILV
Lecture Name	Drive Technology
Semester	5
Season	WS
Lecturer	Daniel T. McGuiness, Ph.D
Module Responsible	DaM
Software	Python, Simulink
SWS Total	3
UE Total	45
ECTS	5
Working Language	English

First Steps

Lecture Information

Title
Advanced electric drives: analysis control and modeling using MATLAB/Simulink
Analysis of electric machinery and drive systems
Design of rotating electrical machines
Electric Machinery Fundamentals (5th Edition)
Electric Machinery
Electric Motors and Drives: Fundamentals Types and Applications
Electrical Machines and Drives: Fundamentals and Advanced Modelling
Electrical machines, drives, and power systems

Table 2: Lecture sources which can be useful during the course of the lecture. For more information on sources, please consult the repo.

First Steps

Lecture Sources

Slide No: 10

- The lecture will have a single personal assignment comprising of a set list of questions and a final exam comprising of all the topics covered in the lecture. Instructions will be given in the assignment.
- For the written exam you are allowed to write your personal equation reference paper, as long as it is a single sheet of A4, double sided and contains no exercise or solutions. Your reference sheet will be collected after the exam.

Assignment Type	Value
Personal Assignment	20
Final Exam	30
Lab Work	50
Sum	100

First Steps

Assignments

Topic	Units	Self Study
Magnetic Circuits and Materials	4	4
Transformers	4	4
Electromechanical Energy Conversion	4	8
Lab 1: Installation of a Drive System	3	6
Rotating Magnetic Fields	2	4
Lab 2: Building a Lathe	4	8
Synchronous Machines	4	8
Poly-phase Induction Drives	4	8
Lab 3: Synchronous and Induction Drives	4	8

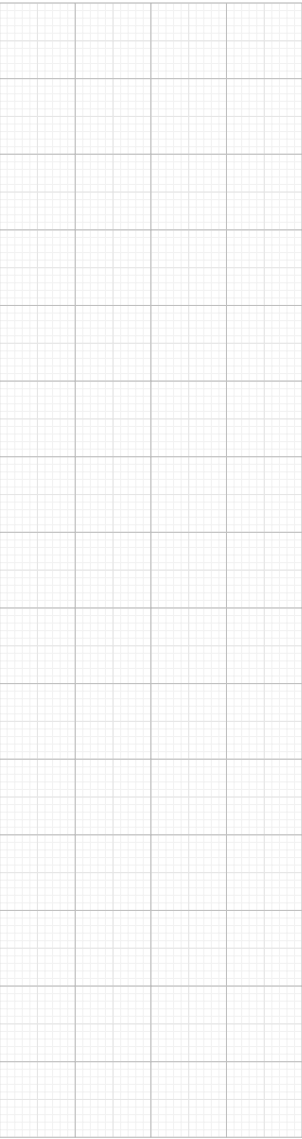
First Steps

Content Preview

Margin Notes

Margin Notes

Slide No: 13



Introduction ii

Topic	Units	Self Study
DC Drives	4	8
Special Machines	2	3
Lab 4: Photovoltaic Cells	4	8
Final Exam	2	4
Total	45	90

First Steps

Content Preview

Table of Contents

Table of Contents

Learning Outcomes

Introduction

Magnetic Materials

Magnetic Circuits

Maxwell's Equations

Flux Linkage, Inductance and Energy

Magnetic Materials

Introduction

Ferromagnetism

B-H Curve

AC Excitation

Definitions

Hysteresis Losses

Bibliography

List of References

Magnetic Circuits and Materials

Slide No: 15

Slide No: 14

Magnetic Circuits and Materials

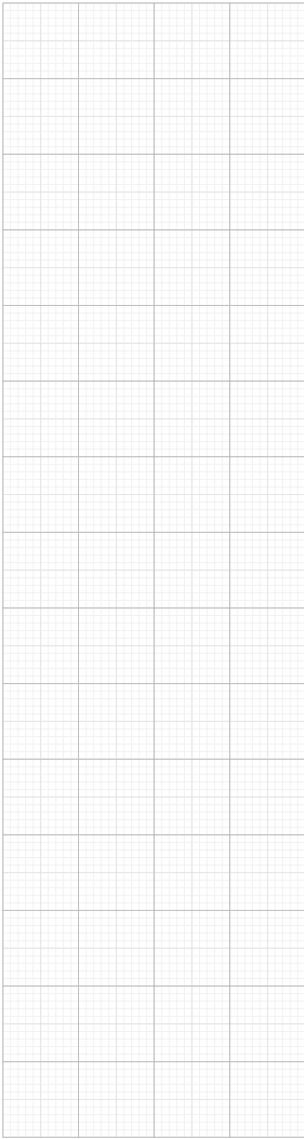
(LO1) An Overview of Maxwell's Equations,
(LO2) Introduction to Magnetic Circuits,
(LO3) Brief look on Magnetic Materials,
(LO4) Magnetic Material Losses.



Learning Outcomes

Slide No: 16

Margin Notes



Margin Notes

Slide No: 17

Magnetic Circuits and Materials

- Almost all electric drives use **ferromagnetic material** for shaping and directing **B**-fields.
- **B**-fields act as the medium for **transferring/converting** energy.

Permanent Magnet (PM)s are also widely used in drive design.

- Without these materials, practical implementations of most familiar electro-mechanical devices would not be possible.
- Analysing and describing systems containing them is essential for designing effective drives.
- We start by looking at Maxwell's equations.

Introduction

Magnetic Materials

Magnetic Circuits and Materials

- Solution to Maxwell's equation can be hard if not attainable and usually simplifying assumptions are made to reach practical solutions.
- First, the displacement-current term ($\epsilon_0 \partial \mathbf{E} / \partial t$) can be neglected.

This term accounts for **B**-fields produced in space by time-varying **E**-fields (i.e., electromagnetic radiation) [2].

- Neglecting $\epsilon_0 \partial \mathbf{E} / \partial t$ results in the magneto-static form which relates **B**-fields to the currents (**J**) which produce them.
- In integral form [3]: 6 / 56

$$\oint_C \mathbf{H} d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a}, \quad \text{and} \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0.$$

Magnetic Circuits

Maxwell's Equations

Slide No: 19

Slide No: 18

Magnetic Circuits and Materials

- Maxwell's equations are a set of coupled partial differential equations which form the foundation of electric and magnetic circuits.
- In their PDE form, they are written as [1]:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right). \end{aligned}$$

where **E** is the electric field, **B** is the magnetic field, ρ is the electric charge density and **J** the current density, ϵ_0 is the vacuum permittivity and μ_0 is the vacuum permeability.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

$$\oint_C \mathbf{H} d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a}, \quad \text{and} \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0.$$

The line integral ($\oint$) of the tangential component of the magnetic field intensity (**H**) around a closed contour (**C**) is equal to the total current (**J**) passing through any surface (**S**) linking that contour.

- The second one states the magnetic flux density (**B**) is conserved, i.e., no net flux enters or leaves a closed surface (i.e., $\nabla \cdot \mathbf{B} = 0$).

These allow us to remove **E** field from our calculations.

Magnetic Circuits

Maxwell's Equations

Slide No: 20

Margin Notes

Margin Notes

Slide No: 21

Magnetic Circuits and Materials

- A second simplifying assumption involves the concept of magnetic circuits.
- The general solution for the H and the B in a structure of complex geometry is extremely difficult if not practically impossible.
- However, a 3D field problem can often be reduced to what is essentially a circuit equivalent of magnetic elements,
 - with acceptable engineering accuracy [4].

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

The diagram shows a rectangular magnetic core with a vertical air-gap on the right side. A coil with N_1 turns is wound around the left vertical leg. The magnetic flux ϕ is shown circulating clockwise through the core. The air-gap length is labeled g . A legend on the right defines the symbols: μ (Magnetic core Permeability), μ_0 (Air-gap Permeability), l_c (Mean Core Length), A_c (Cross-sectional Area), and A_g (Air-gap Cross-sectional Area). The winding is labeled 'Windings N_1 Turns' and the flux path is labeled 'Magnetic Flux Path'.

Magnetic Circuits

Maxwell's Equations

Figure 1: An example of a magnetic circuit.

Slide No: 23

Slide No: 22

Magnetic Circuits and Materials

- A magnetic circuit is a structure composed of highly permeable, $\mu_r \gg 0$ material(s).
- The presence of high-permeability causes magnetic flux to be confined to the paths defined by the structure.

Magnetic Circuits

Maxwell's Equations

This is similar to how currents are confined to the conductors of an electric circuit.

Magnetic Circuits and Materials

Let's have a further look at this setup:

- The core is assumed to be composed of magnetic material with permeability much greater than of surrounding air (i.e., $\mu_r \gg \mu_0$).
- The core is of uniform cross section and is excited by a winding of N turns carrying a current of i amperes.

Magnetic Circuits

Maxwell's Equations

This winding produces a B -field in the core.

Slide No: 24

Margin Notes

Drive Technology

Page 6 of 18

Margin Notes

Slide No: 25

Magnetic Circuits and Materials



- Due to high permeability of the material (i.e., $\mu_r \gg \mu_0$), the magnetic flux is confined **almost entirely to the core**,
- The **B**-field lines follow the path defined by the core,
- The flux density is essentially uniform over a cross section as the cross-sectional area is **uniform**.
- The **B**-field can be visualised in terms of flux lines which form closed loops interlinked with the winding.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials



$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a}$$

- This equation states that all flux entering the surface enclosing a volume must leave the volume as they form closed loops.
- This is enough to justify the uniformity of the magnetic flux density (**B**) across the cross section of a magnetic circuit (A_c):

$$\phi_c = B_c A_c,$$

where ϕ_c is the flux, B_c is the magnetic flux density magnitude and A_c is the cross-sectional core area.

Magnetic Circuits

Maxwell's Equations

Slide No: 27

Slide No: 26

Magnetic Circuits and Materials



- The source of the **B**-field in the core is the ampere-turn Ni .
- In magnetic circuit terminology Ni is the Magneto-motive Force (MMF) acting on the magnetic circuit.
- In systems with more than one winding, Ni must be replaced by the **algebraic sum** of the ampere-turns of all the windings:

$$\sum_{n=0}^{\infty} N_i i_i.$$

- The magnetic flux (ϕ) crossing a surface S is the surface integral of the normal component of **B**:

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a}$$

where the unit of flux (ϕ) is Wb.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials



- Using the original magneto-static simplifications (i.e., no **E**-fields), we can come up to:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}.$$

- As the path of the flux line is close to the mean length¹ of the core (l_c), this is simplified to:

$$\mathcal{F} = Ni = H_c l_c,$$

where H_c is the **average magnitude** of **H**.

Magnetic Circuits

Maxwell's Equations

Slide No: 28

Margin Notes

Margin Notes

Slide No: 29

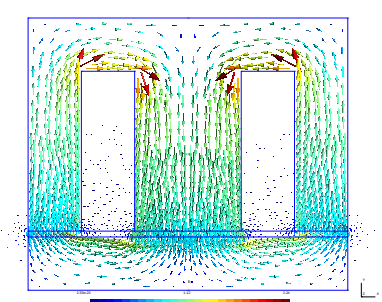
Magnetic Circuits and Materials

- The relationship between the H and B is a **property of the material**.
- It is common to assume a linear relationship shown as:
$$B = \mu H,$$
where μ is the magnetic permeability.
- In SI units, H is measured in $A \cdot m^{-1}$ and B is in $Wb \cdot m^{-2}$ or T.

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials



Magnetic Circuits

Maxwell's Equations

Figure 2: A Magnetic Circuits with two mmf sources.

Slide No: 30

Magnetic Circuits and Materials

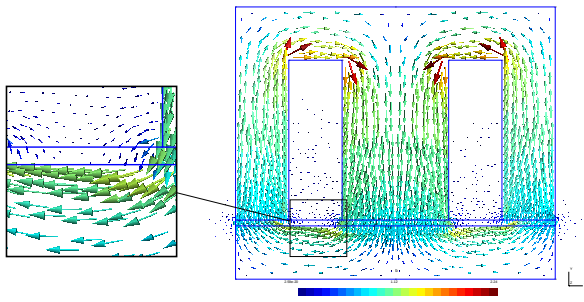
- Transformers are wound on **closed cores**.
- However, electro-mechanical devices which incorporate a **moving element** must have **air gaps** in their magnetic circuits.
- When the air-gap (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

Magnetic Circuits

Maxwell's Equations

If g becomes excessively large, the flux will be leak out of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable.

Magnetic Circuits and Materials



Magnetic Circuits

Maxwell's Equations

Figure 3: A closeup of the fringe paths caused by the air gap.

Margin Notes

Drive Technology

Page 8 of 18

Margin Notes

Slide No: 33

Magnetic Circuits and Materials

- Provided the air-gap length g is sufficiently small, the configuration can be analysed as a magnetic circuit with two (2) series components:

- Magnetic core (μ), cross-sectional area A_c , and mean length l_c ,
- Air gap (μ_0), cross-sectional area A_g , and length l_g .

- In the core the flux density can be assumed uniform and ($\mu \gg \mu_0$):

$$B_c = \frac{\phi}{A_c}, \quad \text{and in the air gap, } B_g = \frac{\phi}{A_g},$$

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

- The term which multiplies mmf to flux is called **Reluctance**.

$$\mathcal{F} = \phi \left(\underbrace{\frac{l_c}{\mu A_c}}_{\mathcal{R}_c} + \underbrace{\frac{g}{\mu_0 A_g}}_{\mathcal{R}_g} \right) \quad \text{or} \quad \mathcal{F} = \phi (\mathcal{R}_c + \mathcal{R}_g)$$

Reluctance can be seen as analogous to resistance to electrical circuits.

Magnetic Circuits

Maxwell's Equations

Slide No: 35

Slide No: 34

Magnetic Circuits and Materials

- Applying this to the magnetic circuit gives:

$$\mathcal{F} = H_c l_c + H_g g,$$

using the **B-H** relationship:

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g.$$

- Here, $\mathcal{F}$ is the MMF applied to the magnetic circuit where $\mathcal{F}_c = H_c l_c$ is the magnetic field in the core and $\mathcal{F}_g = H_g l_g$ is the magnetic field in the air-gap.

- Using ϕ we can re-write this equation as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right).$$

Magnetic Circuits

Maxwell's Equations

Magnetic Circuits and Materials

- When a **B**-field varies with time, an **E**-field is produced in space as determined by Faraday's law [1]:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

The line integral of the electric field intensity **E** around a closed contour C is equal to the time rate of change of the magnetic flux linking (i.e. passing through) that contour.

In electric drives, the effects of **E** field can be neglected.

Flux Linkage, Inductance and Energy

Slide No: 36

Margin Notes

Margin Notes

Slide No: 37

Magnetic Circuits and Materials

- This simplification allows us to cancel the LHS of the equation to just the induced voltage (i.e., electromotive force)

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

- Whereas on the RHS, the flux is dominated by the core flux (ϕ).

$$\mathcal{E} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- For a magnetic circuit composed of magnetic material of constant magnetic permeability, the relationship between ϕ and i will be linear and we can define the inductance L as:

$$L = \frac{\lambda}{i} \text{ which can be written as } L = \frac{N^2}{\mathcal{R}_{\text{tot}}}$$

where L is the inductance measured in Henry (H).

Inductance requires a linear relationship between flux and mmf.

Flux Linkage, Inductance and Energy

Slide No: 39

Slide No: 38

Magnetic Circuits and Materials

- As winding links the core flux N times, our equation is simplified to:

$$\mathcal{E} = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \text{ where } \lambda = N\phi$$

where λ is flux linkage, measured in (Wb · Turn),

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

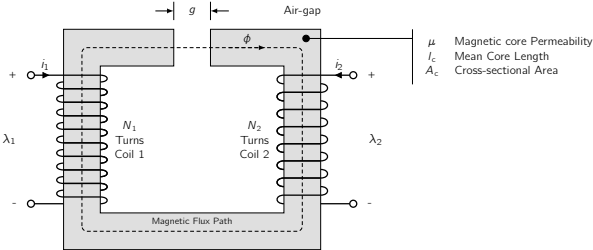


Figure 4: An example of a magnetic winding circuit with 2 windings.

Flux Linkage, Inductance and Energy

Slide No: 40

Margin Notes

Margin Notes

Slide No: 41

Magnetic Circuits and Materials

- The MMF ($\mathcal{F}$) is given by the total ampere-turns.
- The reference directions for the currents (i_1, i_2) have been chosen to produce flux in the same direction.
- The total mmf is therefore:

$$\mathcal{F} = N_1 i_1 + N_2 i_2.$$

- With the reluctance of the core neglected and assuming that $A_c = A_g$, the core flux ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g},$$

where ϕ is the **resultant core flux** produced by the total mmf of the two windings.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- Similarly, the flux linkage on coil 2 is:

$$\lambda_2 = N_2 \phi = N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_1^2 \left(\frac{\mu_0 A_c}{g} \right) i_2$$

and this is simplified to:

$$\lambda_1 = L_{21} i_1 + L_{22} i_2$$

where $L_{21} = L_{12}$ is mutual inductance and L_{22} is self inductance.

It is important to note the ϕ is calculated based on the superposition of i_1, i_2 which implies a linear relationship between flux and mmf.

Flux Linkage, Inductance and Energy

Slide No: 43

Slide No: 42

Magnetic Circuits and Materials

- This resultant ϕ determines the operating point of the core material.
- If ϕ is broken up into terms attributable to the individual currents, the resultant flux linkages of coil 1 can be expressed as:

$$\lambda_1 = N_1 \phi = N_1^2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_2,$$

and this is simplified to:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2,$$

where L_{11} is the **self** inductance and L_{12} is the **mutual** inductance of the coil.

Flux Linkage, Inductance and Energy

Magnetic Circuits and Materials

- It cannot be rigorously applied in situations where the **nonlinear characteristics of magnetic materials** dominate.
- However, in many situations of practical interest, the reluctance of the system is dominated by that of an air gap (which is of course linear) and the nonlinear effects of the magnetic material can be ignored.
- In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material.
- Using this it is possible to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.

Flux Linkage, Inductance and Energy

Slide No: 44

Margin Notes

Slide No: 45

Magnetic Circuits and Materials



- Electro-mechanical devices require two (2) important properties:
 - Obtain large magnetic flux densities ($\uparrow \mathbf{B}$)
 - Requiring relatively low levels of magnetizing force ($\downarrow \mathbf{H}$).

Since magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.

Magnetic Materials

Introduction

Magnetic Circuits and Materials




Figure 5: The periodic table of elements. Of all the elements that exist only a few of them have the ability to generate a magnetic field without any external need. The elements which exhibit ferromagnetism has been highlighted.

Magnetic Materials

Ferromagnetism

Slide No: 47

Slide No: 46

Magnetic Circuits and Materials



- Magnetic materials can be used to **constrain and direct B -fields** in well-defined paths.
- In a transformer, they are used to maximize the coupling between the windings as well as to **lower the excitation current** required for transformer operation.
- In electric drives, they are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

Magnetic Materials

Introduction

Magnetic C

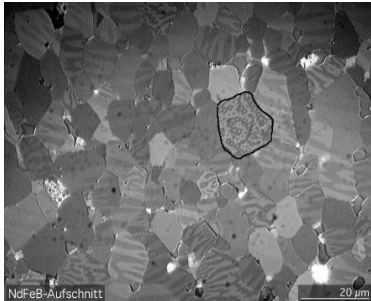
- Typically composed of iron (**Fe**) and alloys of iron with:
 - Cobalt, Tungsten, Nickel, Aluminium, and other metals
- These materials are characterised by a wide range of properties,
 - Phenomena responsible for their properties **are the same**.
- Ferro-magnetic materials are found to be composed of a large number of **domains**.
 - Regions where the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain.
- In an un-magnetised sample of material, the domain magnetic moments are **randomly** oriented, and the net resulting magnetic flux in the material is zero (**0**).

Slide No: 48

Margin Notes

Slide No: 49

Magnetic Circuits and Materials



The micrograph shows a polycrystalline structure of Nd₂Fe₁₄B. The grains are roughly polygonal and vary in size. Within each grain, there are alternating light and dark regions representing magnetic domains. One grain in the center-right is outlined with a black line. The text 'NdFeB-Aufschnitt' is in the bottom left, and '20 μm' is in the bottom right.

Figure 6: Microcrystalline grains within a piece of Nd₂Fe₁₄B (the alloy used in neodymium magnets) with magnetic domains made visible with a Kerr microscope. The domains are the light and dark stripes visible within each grain. The outlined grain has its magnetocrystalline axis almost vertical, so the domains are seen end-on [5].

Magnetic Materials

Ferromagnetism

Magnetic Circuits and Materials



- Without an external **B** -field, magnetic domains naturally align along certain directions associated with the **crystal structure** of the domain.
 - This is known as axes of easy magnetisation [6].
- If the applied **B** is reduced, the domain magnetic moments relax to the direction of easy magnetism nearest to the applied field.
- As a result, **B** is reduced to zero (0), although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation;
- They will retain a net magnetisation **along the applied** field.
- It is this effect which is responsible for the phenomenon known as magnetic hysteresis [7].

Magnetic Materials

Ferromagnetism

Slide No: 51

Slide No: 50

Magnetic Circuits and Materials



- When ***B***-field is applied to the material, the magnetic domain moments tend to align with the applied ***B***-field.
- As a result, the magnetic moments add to the ***B***-field, producing a higher flux density than would exist due to the magnetising force alone.
- Therefore the effective permeability μ , equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0).
- As the magnetising force (***H***) is increased, this continues until all the magnetic moments are aligned with the applied field.
 - at this point they can no longer contribute to increasing the magnetic flux density

Material becomes fully saturated.

Magnetic Materials

Ferromagnetism

Magnetic Circuits and Materials

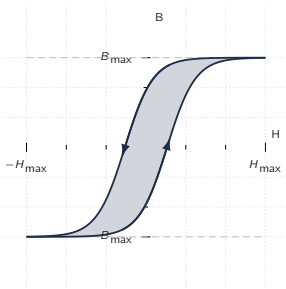


Figure 7: An example of an B-H Hysteresis curve.

Magnetic Materials

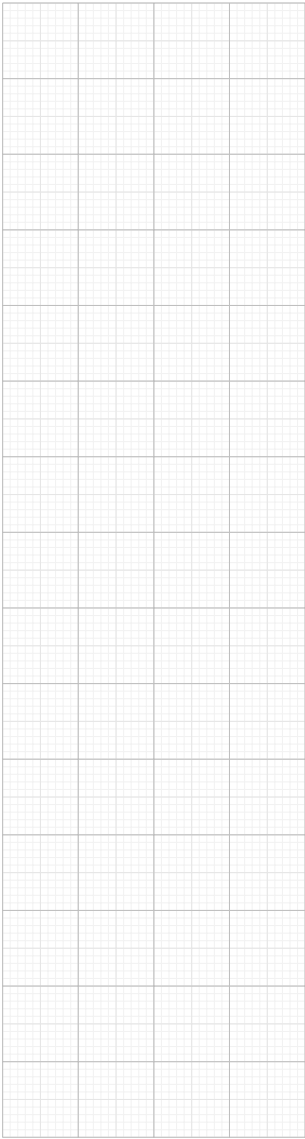
B-H Curve

Slide No: 52

Margin Notes

Margin Notes

Slide No: 53



Magnetic Circuits and Materials

1st Quadrant

- The domains are very small, below the single domain size where there is resistance to demagnetization [8].
- The domains start to increase in size.

For a small interval, the magnetization will be reversible.

- As the field increases, magnetization will no longer reverse to zero but move on a minor hysteresis loop.
- Eventually the curve starts to bend over to the right.
 - It will still increase as more magnetic domains reach their full size and their magnetisation become parallel to the external field [9].

Magnetic Materials

B-H Curve

Magnetic Circuits and Materials

2nd Quadrant - Demagnetisation

- In this quadrant the applied field **opposes** the materials **B**-field.
- As the external **B**-field increases in magnitude, some domains will reverse.
- At the knee of the demagnetisation curve, this increase has become rapid and the magnetisation will fall to the H_{ci} point.
- At H_{ci} , the number of domains aligned with the original magnetization is the same as the number aligned with the opposing magnetic field.

The net magnetisation is zero (0).

Magnetic Materials

B-H Curve

Slide No: 55

Slide No: 54

Magnetic Circuits and Materials

1st Quadrant - Saturation

- Eventually the **B**-field will become high enough where no more change in the magnetisation occurs.
 - This is called **technical saturation**.
- It is possible to reach 99+% saturation.
- As the field is backed off from "saturation", the magnetisation declines very slightly to the B_r point.
 - This is the remanence, or residual induction [10].

All of the magnetic energy is now in the magnet and its field.

Magnetic Materials

B-H Curve

Magnetic Circuits and Materials

3rd Quadrant - Re-magnetisation

- The total magnetization of the part will be **reversed**.
- If we go far enough, magnetization will reach the saturation level in the negative direction.

4th Quadrant - Demagnetisation

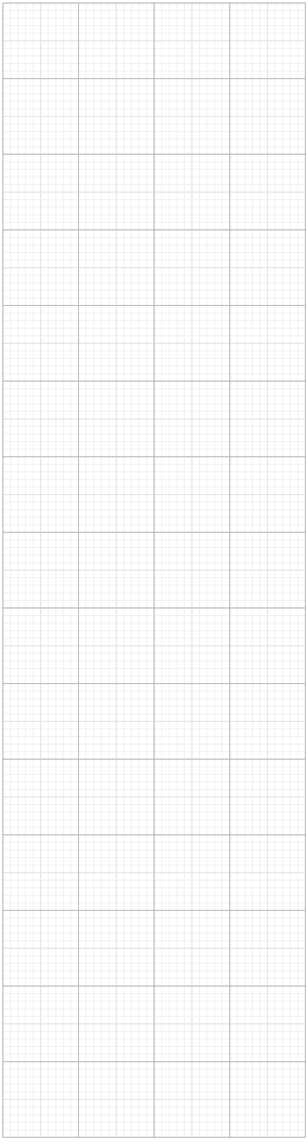
- After fully reversing the magnet and removing the field in the third quadrant, magnetization will recoil to a point that is the negative of the B_r observed when in the first and second quadrant.
- If we apply additional field in the positive direction, we duplicate the second quadrant curve.

Magnetic Materials

B-H Curve

Slide No: 56

Margin Notes



Margin Notes

Slide No: 57

Magnetic Circuits and Materials

- In AC power systems, voltage and flux wave-forms closely approximate sinusoidal functions over time.
- Time to describe the excitation characteristics and losses associated with steady-state ac operation of magnetic materials.
- Assume a closed-core magnetic circuit (i.e., with no air gap),
- And a sinusoidal variation of the core flux (ϕ) with the following:

$$\phi(t) = \phi_{\max} \sin(\omega t) = A_c B_{\max} \sin(\omega t)$$

where:

$\phi_{\max}$ amplitude of core flux in Wb,
 $B_{\max}$ Amplitude of flux density B_c in T,
 ω Angular frequency in rad/s,
 f Frequency in Hz.

AC ExcitationDefinitions

Magnetic Circuits and Materials

- The rms value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore the induced voltage rms is:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N A_c B_{\max} = \sqrt{2} \pi N A_c B_{\max} = 4.44 N A_c B_{\max}$$

- To produce magnetic flux in the core requires current in the exciting winding known as the exciting current, i_ϕ .

The nonlinear magnetic properties of the core require that the wave-form of the exciting current differs from the sinusoidal waveform of the flux.

- When the hysteresis curve saturates the excitation current spikes.

AC ExcitationDefinitions

Slide No: 59

Slide No: 58

Magnetic Circuits and Materials

- From, the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{\max} \cos(\omega t) = E_{\max} \cos(\omega t)$$

where:

$$E_{\max} = \omega N \phi_{\max} = 2\pi f N A_c B_{\max}$$

- In steady-state AC, it is more important to use rms rather than instant values.
- Generally, the rms value of a periodic function of time $f(t)$, of T is:

$$F_{\text{rms}} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt\right)}$$

AC ExcitationDefinitions

Magnetic Circuits and Materials

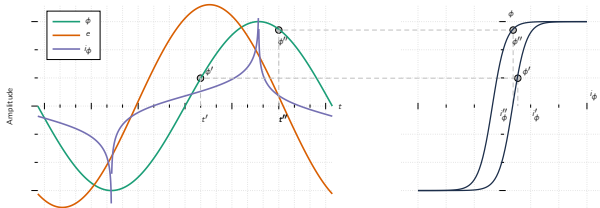


Figure 8: An Example Visualisation of the excitation current experienced by a magnetic circuit.

AC ExcitationDefinitions

Slide No: 60

Margin Notes

Slide No: 61

Magnetic Circuits and Materials



- The exciting current supplies the mmf required to produce the core flux and the power input associated with the energy in the magnetic field in the core.
- Part of this energy is dissipated as losses and results in heating of the core.
- The rest appears as reactive power associated with energy storage in the magnetic field.

This reactive power is not dissipated in the core as it is cyclically supplied and absorbed by the excitation source.

AC Excitation

Definitions

Magnetic Circuits and Materials



- To counteract the corresponding demagnetizing effect by the **eddy currents**, the current in the exciting winding must increase.

Therefore the resultant "dynamic" B-H loop under ac operation is somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

- It is for this reason that the characteristics of electrical steels vary with frequency and hence manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.

AC Excitation

Definitions

Slide No: 63

Slide No: 62

Magnetic Circuits and Materials



- There are two (2) mechanisms associated with time varying **B**-fields.

Ohmic Heating

- Associated with induced currents in the core material.
- From Faraday's law, we see that **B**-fields give rise to **E**-fields.
- In magnetic materials these **E**-fields result in induced currents,
 - These are known as **eddy currents**, circulating in the core material, opposing the material's internal **B**-field [11].

AC Excitation

Definitions

Magnetic Circuits and Materials



- To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.
- These laminations, which are aligned in the direction of the field lines, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish.
- This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths; the thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the square of the excitation frequency and also as the square of the peak flux density.

AC Excitation

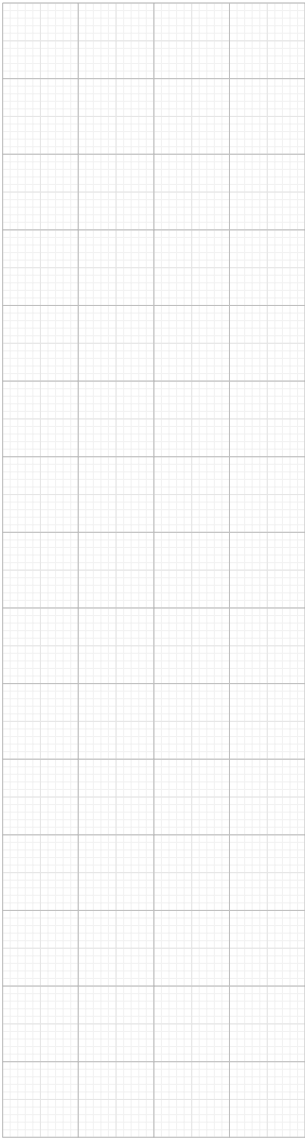
Definitions

Slide No: 64

Margin Notes

Margin Notes

Slide No: 65



Magnetic Circuits and Materials

- Due to the **hysteretic nature** of magnetic material.
- In a magnetic circuit or the transformer, a time-varying excitation (i_ϕ) will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop.
- The energy input W to the magnetic core as the material undergoes a single cycle is shown to be:
$$W = \oint i_\phi d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c$$
- Notice $A_c l_c$ is the **core volume** and the integral is the area of the ac hysteresis loop, we see that each time the magnetic material undergoes a cycle, there is a net energy input into the material.

AC Excitation

Hysteresis Losses

Magnetic Circuits and Materials

- In general, both losses depend on the metallurgy of the material as well as the flux density and frequency.
- Information on core loss is typically presented in graphical form.
- It is plotted in terms of watts per unit weight as a function of flux density;
 - Often a family of curves for different frequencies are given.
 - Generally it is either 50 or 60 Hz.

AC Excitation

Hysteresis Losses

Slide No: 67

Slide No: 66

Magnetic Circuits and Materials

- This is the required energy to move around the magnetic dipoles in the material and is dissipated as heat in the material.
- Therefore for a given flux level, the corresponding hysteresis losses are **proportional to the area of the hysteresis loop** and to the **total volume of material**.
- As there is an energy loss per cycle, hysteresis power loss is **proportional to the frequency of the applied excitation**. (Data Sheet)

AC Excitation

Hysteresis Losses

Magnetic Circuits and Materials

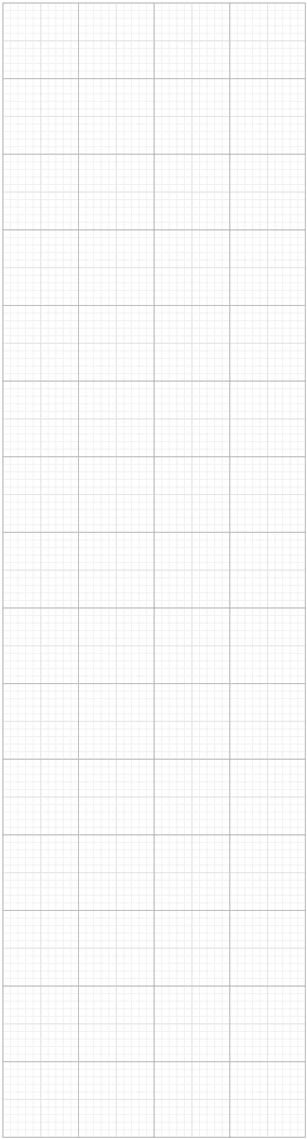
- Nearly all transformers and certain sections of electric machines use sheet-steel material that has highly favorable directions of magnetization along which the core loss is low and the permeability is high.
- This material is termed **grain-oriented steel** [12].
- The reason lies in the atomic structure of a crystal of the silicon-iron alloy,
 - which is a body-centred cube.

AC Excitation

Hysteresis Losses

Slide No: 68

Margin Notes



Slide No: 69

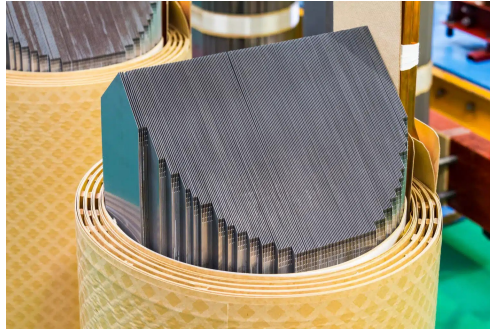


Figure 9: An example of Grain Oriented Steel used in industry [13].

AC Excitation

Hysteresis Loses

List of References i

- [1] David J Griffiths. *Introduction to electrodynamics*. Cambridge University Press, 2023.
- [2] Hsueh-Yuan Pao, Steven L Dvork, and Donald G Dudley. "The effects of neglecting displacement currents when studying transient wave propagation in the Earth". In: *IEEE Transactions on Antennas and Propagation* 44.9 (1996), pp. 1259–1265.
- [3] Richard P. Feynman. *Feynman lectures on physics. Volume 2: Mainly electromagnetism and matter*. 1964.
- [4] MIT Staff. *Magnetic Circuits and Transformers: A First Course for Power and Communication Engineers*. MIT Press, 1943.
- [5] Gorchy. *Photomicrograph of NdFeB*. 2005. URL: <https://en.wikipedia.org/wiki/File:NdFeB-Domains.jpg>.
- [6] Hiroaki Mamiya and Balachandran Jayadevan. "Design criteria of thermal seeds for magnetic fluid hyperthermia-from magnetic physics point of view". In: *Nanomaterials for magnetic and optical hyperthermia applications*. Elsevier, 2019, pp. 13–39.
- [7] Peter J Wasilewski. "Magnetic hysteresis in natural materials". In: *Earth and Planetary Science Letters* 20.1 (1973), pp. 67–72.
- [8] William Fuller Brown Jr. "Rigorous approach to the theory of ferromagnetic microstructure". In: *Journal of Applied Physics* 29.3 (1958), pp. 470–471.

Bibliography

List of References

Slide No: 71

Slide No: 70

List of Acronyms i

MMF The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.. 26, 34, 41

PM An object made from a material that is magnetized and creates its own persistent magnetic field. 17

AC Excitation

Hysteresis Loses

List of References ii

- [9] Alliance Org. "Understanding the Hysteresis (B-H) Curve". In: (2024).
- [10] Terunobu Miyazaki and Hannin Jin. *The physics of ferromagnetism*. Vol. 158. Springer Science & Business Media, 2012.
- [11] Fausto Fiorillo. *Measurement and characterization of magnetic materials (electromagnetism)*. Elsevier Academic Press, 2005.
- [12] Colin Tong. *Introduction to materials for advanced energy systems*. Springer, 2019.
- [13] Metal Miner. *Grain Oriented Electrical Steel Demand Expected to Keep Growing*. 2024. URL: <https://agmetalmminer.com/2023/01/30/grain-oriented-electrical-steel-demand-expected-to-keep-growing/>.

Bibliography

List of References

Slide No: 72

Margin Notes